Open Spatial Computing Platform

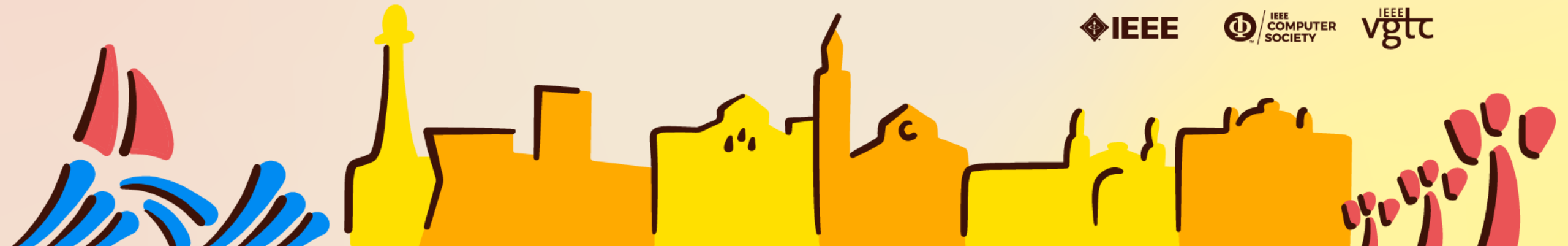
Gábor Sörös

Open AR Cloud / Nokia Bell Labs

ISMAR 2026

October 5 - 9, Bari, Italy

The 25th International Symposium on Mixed and Augmented Reality



Vision: Ubiquitous, geographically anchored digital information

Location-based AR becomes mainstream



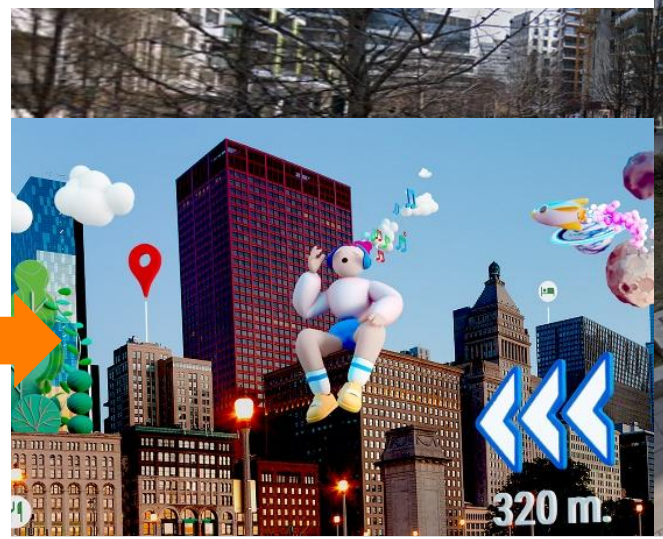
Niantic Pokémon GO, 2016



Spatial Internet, 2018
(concept by Judith Hecktor)

NIANTIC SPATIAL

Niantic Spatial, 2025



Google Geospatial AR, 2024



Google Street View at Corvin Budapest, 2022

Such physical hyperlinks make our world clickable!

Spatial anchors and shared maps of the world

Required for re-identification of physical locations

Anchoring 3D virtual assets to the real world requires precise and robust localization (<5cm, <1 deg)

Visual localization provides this accuracy and enables recognizing the same location over time and across users

Spatial anchor = localization + discovery + persistence

Shared world map and shared contents stored in the “AR Cloud”

Key players: Google, Apple, Meta, Niantic, (Microsoft), (Nokia)

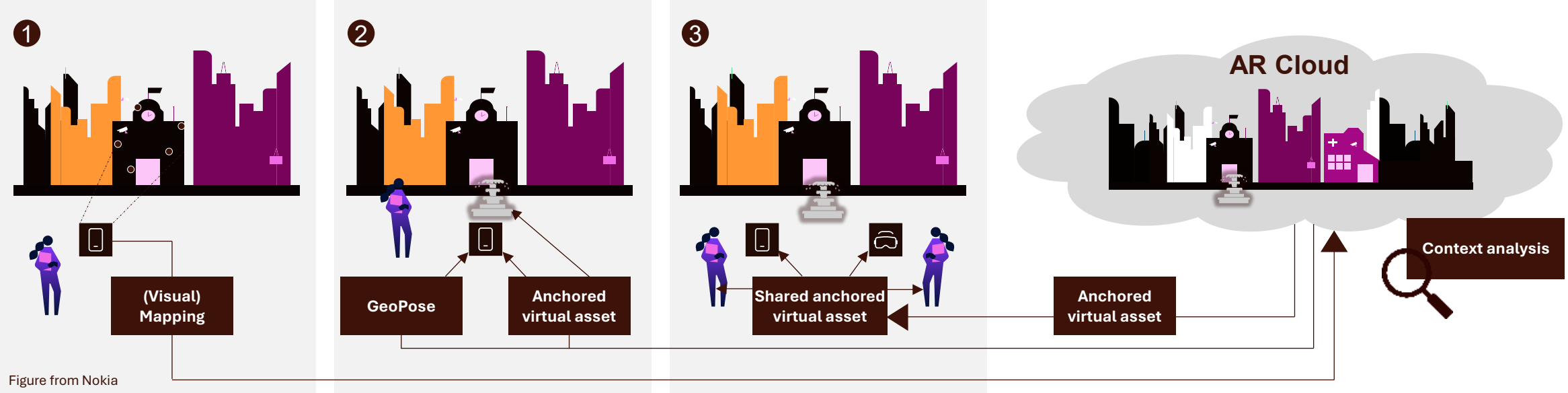
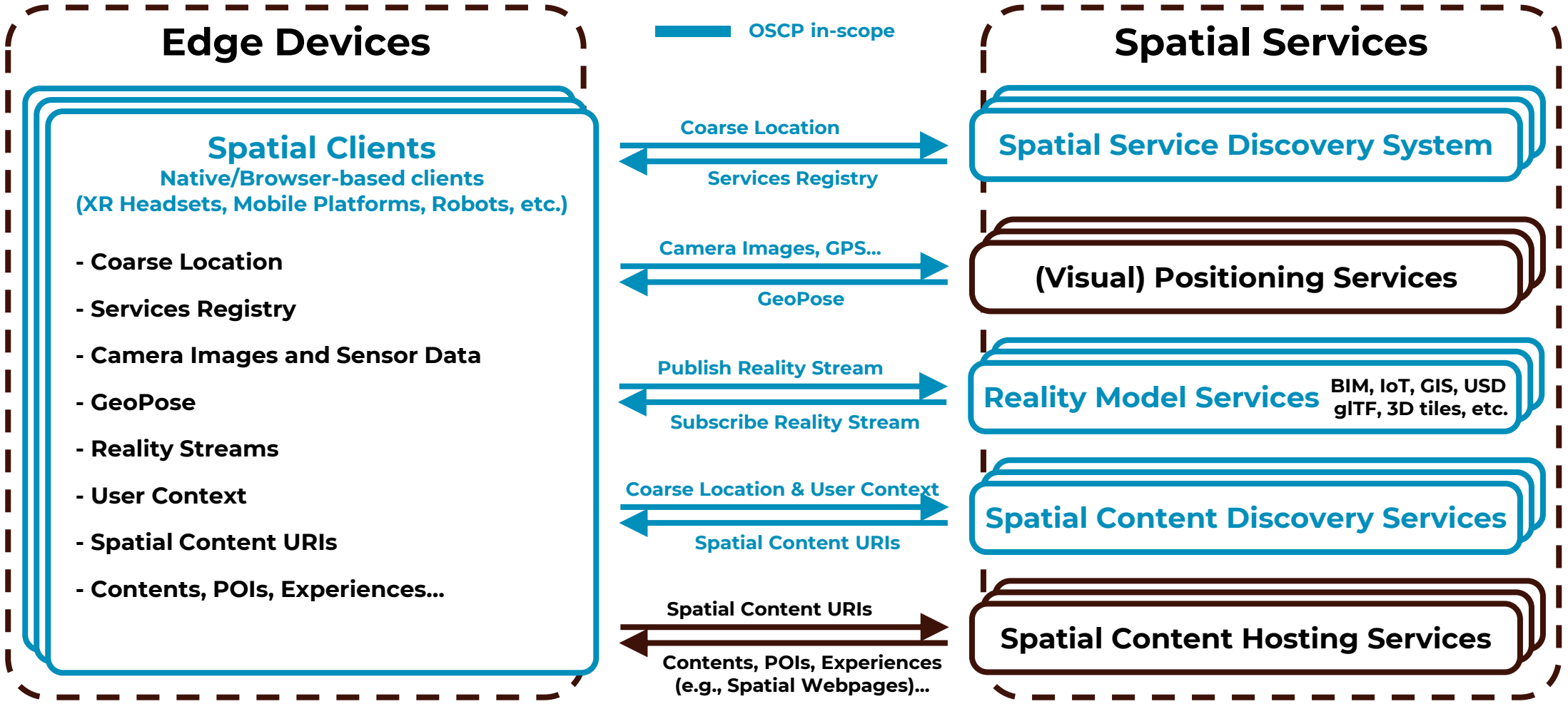


Figure from Nokia

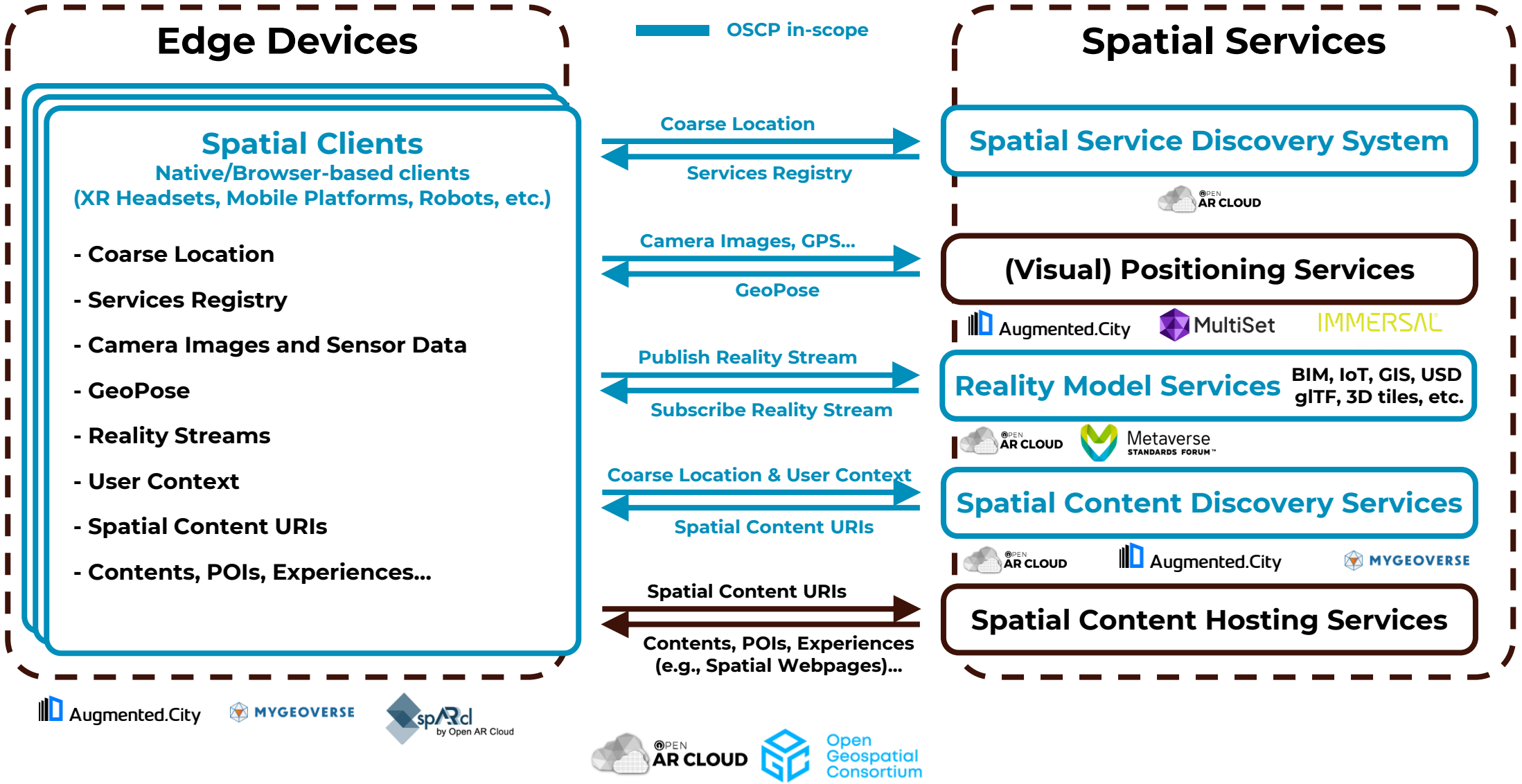
Open Spatial Computing Platform

OSCP Overview

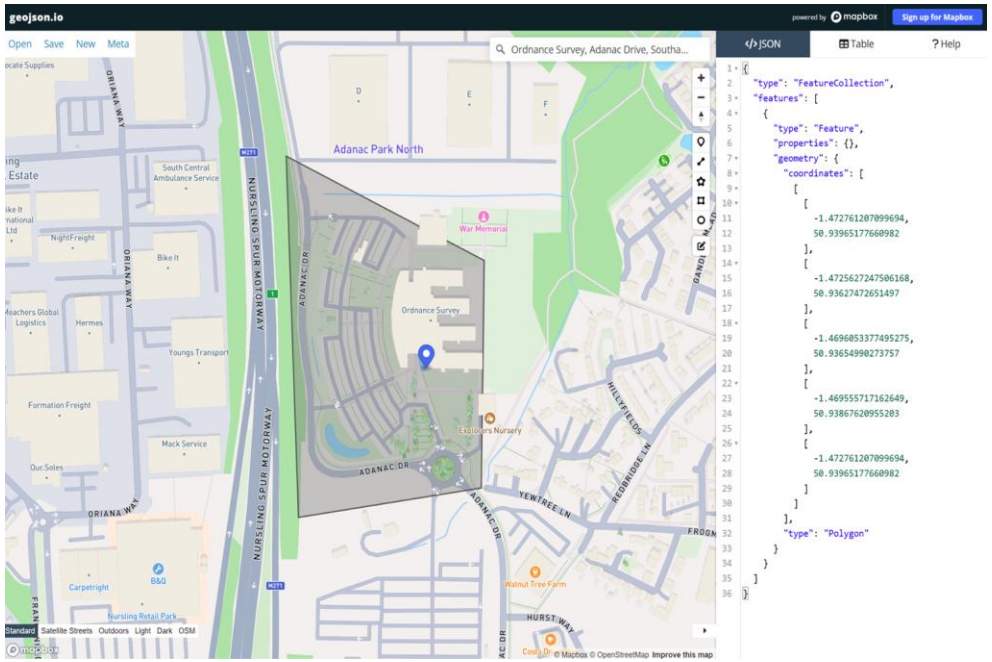


Open Spatial Computing Platform

OSCP Partner solutions



Spatial Discovery – Preview



OSCP Spatial Discovery Gen 1 - HTTP, P2P, OSM

https://github.com/OpenArCloud/oscp-spatial-service-discovery
https://github.com/OpenArCloud/oscp-spatial-content-discovery

OSCP Spatial Discovery Gen 2 - HTTP, no P2P, Tile38

https://github.com/OpenArCloud/SSD-FastAPI-Tile38
https://github.com/OpenArCloud/SCD-FastAPI-Tile38

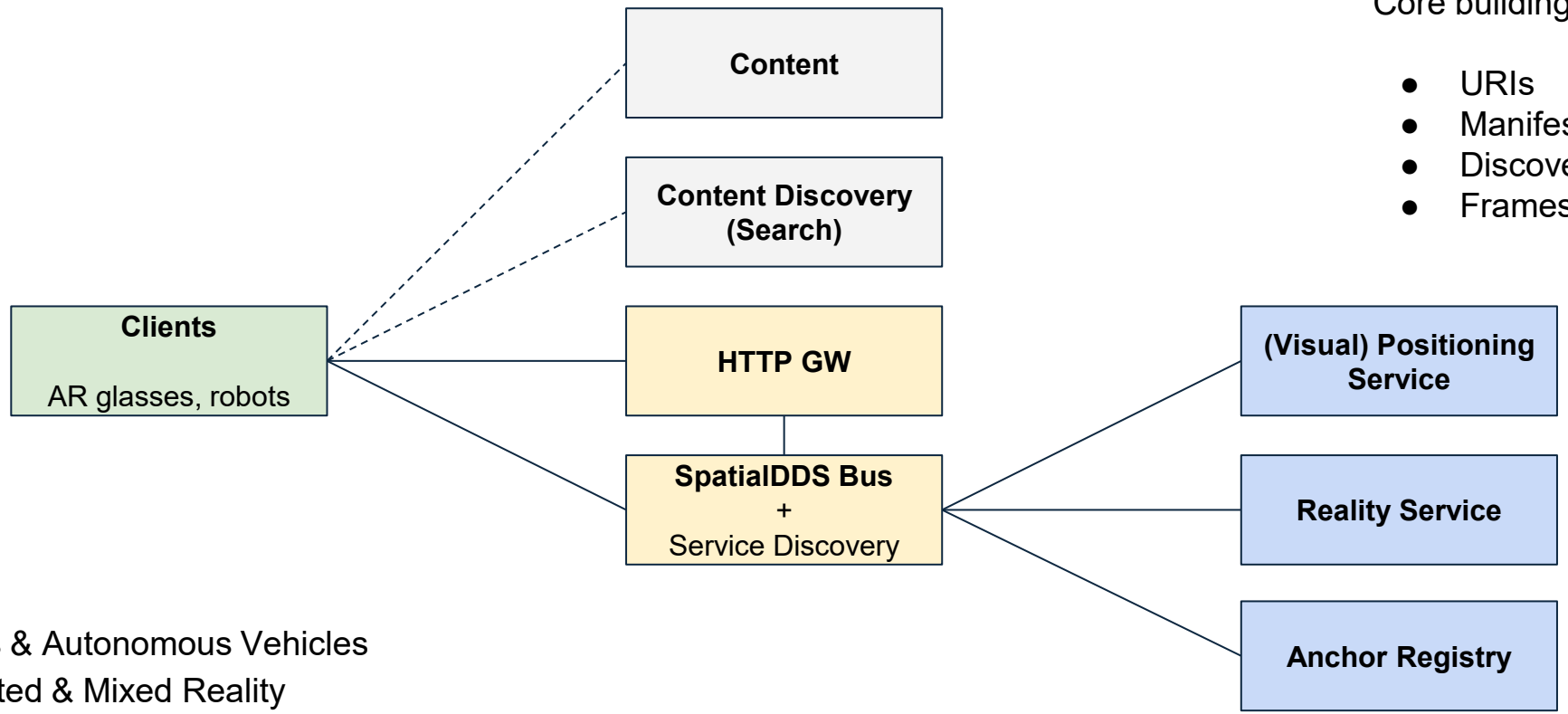
Spatial Content Record (SCR)

```
{
   "contenttype": "model/gltf+json",
   "description": "test model",
   "geopose": {
      "position": {
         "h": 1000.0,
         "lat": 50.93965177660982,
         "lon": -1.472761207099694
      },
      "quaternion": {
         "w": 0.0,
         "x": 0.0,
         "y": 0.0,
         "z": 0.0
      }
   },
   "metatype": "model3D",
   "tags": {
      "name1": "value1",
      "name2": "value2"
   },
   "url": "http://path_to_model"
  }
 ],
 "id": "uuid1"
}
```

Spatial Service Record (SSR)

```
"properties": {
   "id": "uuid1",
   "services": [
      {
         "tags": {
            "name1": "value1",
            "name2": "value2"
         },
         "type": "geopose",
         "url": "http://www.geopose.org"
      }
   ]
}
```

SpatialDDS (Data Distribution Service) - Preview



Core building blocks:

- URIs
- Manifests
- Discovery
- Frames/Transforms

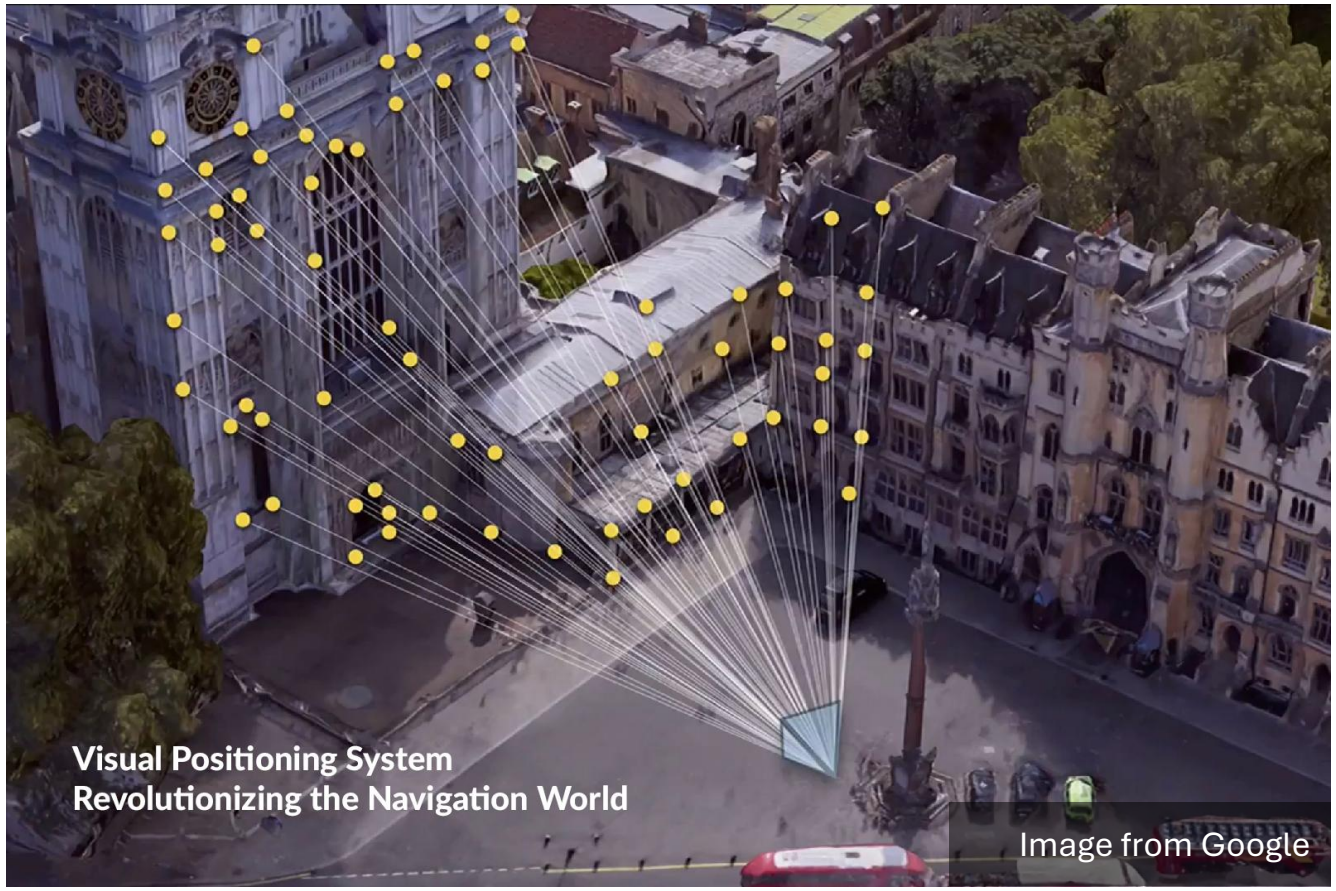
Domains:

- Robotics & Autonomous Vehicles
- Augmented & Mixed Reality
- Digital Twins & Smart Cities
- IoT & Edge AI
- AI World Models & Agents

Services
(also communicate among themselves via SpatialDDS)

Visual positioning systems (VPS) - Preview

Camera-based precise 6DoF localization



Key players:

Google Visual Positioning
Apple GeoAnchors
Facebook (Scape, Mapillary, etc.)
Niantic (6D.ai)
(Microsoft Azure Spatial Anchors)
Augmented City
Immersal
MultiSet

Scalable localization on mobile devices

Local visual-inertial odometry (VIO) + global visual positioning service (VPS)

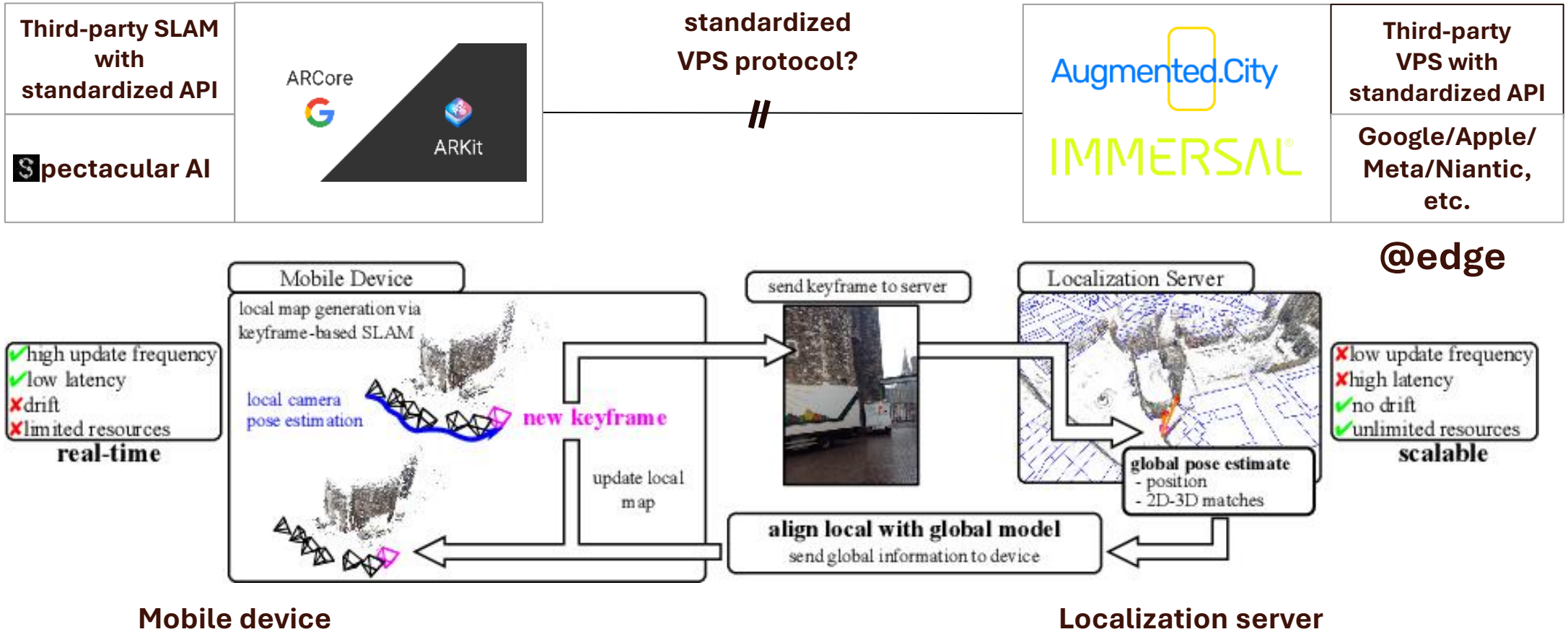


Figure adapted from Middelberg, Sattler, Untzelmann, Kobbelt - Scalable 6-DoF Localization on Mobile Devices, ECCV 2014



Next: GeoPose